

Physical-Space Scarring in Generic Bunimovich Stadia

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Abstract

Motivated by Tao's question on physical-space scarring for the Bunimovich stadium, we prove a physical-space strengthening of Hassell's non-QUE theorem for generic stadia. For the one-parameter family of Dirichlet stadia S_t whose rectangular part has height π and half-length $\pi t/2$, $1 \leq t \leq 2$, we show that for Lebesgue almost every t there are real eigenfunctions u_j and a smooth mean-zero physical observable Q such that $\langle Qu_j, u_j \rangle$ has a non-zero subsequential limit. Consequently, after passing to the same subsequence, the mass fails to equidistribute on some fixed region with smooth interior relative boundary. Thus we answer Tao's question affirmatively in Hassell's generic setting.

The proof is based on the two alternatives in Hassell's argument. In the first, a semi-classical measure supported on bouncing-ball covectors is detected by an observable in the caps. In the bounded-window alternative, Hassell gives eigenfunctions with $|\lambda_j - n_j^2| = O(1)$ and fixed positive overlap with a bouncing-ball quasimode $v_n(x, y) = \chi(x)\varphi_n(y)$. This overlap already forces physical scarring: the selected vertical Fourier coefficient has bounded horizontal frequency and non-zero mass, while lower vertical modes average out and higher vertical modes are evanescent in the rectangle.

1 Introduction, statement, and notation

For $1 \leq t \leq 2$, let S_t be the Bunimovich stadium obtained by adjoining two semicircular caps of radius $\pi/2$ to

$$R_t = I_t \times (-\pi/2, \pi/2), \quad I_t = (-\pi t/2, \pi t/2).$$

Let

$$H_t = -\Delta_{S_t}^D$$

be the Dirichlet Laplacian. We use the sign convention $H_t u = \lambda u$, $\lambda > 0$. By smooth functions on $\overline{S}_t$ we mean restrictions to $\overline{S}_t$ of smooth functions defined in an open neighbourhood of $\overline{S}_t$.

A normalized eigenfunction sequence u_j is *physically equidistributed* if, for every continuous physical observable Q on $\overline{S}_t$,

$$\int_{S_t} Q(z) |u_j(z)|^2 dz \longrightarrow \frac{1}{|S_t|} \int_{S_t} Q(z) dz.$$

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It is *physically scarred* if this convergence fails for at least one fixed physical observable. For regions whose boundary has area zero, the corresponding mass formulation is equivalent; in the theorem below the witnessing region will have smooth interior relative boundary. In that region formulation, “smooth boundary” means smooth relative boundary in the interior of the billiard domain; possible intersections with the billiard boundary are irrelevant for area measure.

The stadium is a central example in quantum chaos: its billiard flow is ergodic, as shown by Bunimovich [1], but it has singular invariant measures carried by the vertical bouncing-ball trajectories. The quantum ergodicity theorem of Shnirelman, Colin de Verdiere, and Zelditch [8, 2, 11], together with its billiard version due to Gerard and Leichtnam [3], gives equidistribution for a density-one subsequence of eigenfunctions. The stronger QUE problem asks whether exceptional subsequences can be excluded; see, for example, Rudnick and Sarnak’s formulation in negative curvature [7]. In the stadium, the bouncing-ball quasimodes behind the physical and numerical scarring picture go back to Heller’s work [5] and to the Heller–Zelditch quasimode mechanism [12].

Tao formulated the physical-space scarring problem for the Bunimovich stadium as an open question [9]. After Hassell proved phase-space non-QUE for generic stadia [4], Tao noted that upgrading the conclusion to physical-space non-equidistribution remained a natural direction [10]. The point of the present note is that Hassell’s bounded-window output already contains enough information to make this physical-space extraction.

Theorem 1.1 (Physical-space non-equidistribution for generic stadia). *For Lebesgue almost every $t \in [1, 2]$ there exist normalized real Dirichlet eigenfunctions*

$$H_t u_j = \lambda_j u_j, \quad \|u_j\|_{L^2(S_t)} = 1, \quad \lambda_j \rightarrow \infty,$$

and a real-valued $Q \in C^\infty(\overline{S}_t)$ such that

$$\int_{S_t} Q(z) dz = 0$$

and, after passing to a subsequence,

$$\int_{S_t} Q(z) |u_j(z)|^2 dz \longrightarrow L \neq 0.$$

Consequently, along the same subsequence, there is a physical region $A \subset S_t$, whose interior relative boundary may be chosen smooth, such that

$$\int_A |u_j(z)|^2 dz \not\rightarrow \frac{|A|}{|S_t|}.$$

The external input is Hassell’s proof of non-QUE for generic stadia [4]. We use it as a black box through the two consequences recorded in Proposition 2.1. The later note by Mangoubi and Weller Weiser [6] clarifies the boundary singularity issue in the Z_1 semiclassical-measure part of Hassell’s stadium argument; no such clarification is needed for the purely local bounded-window extraction in Proposition 4.1.

2 The input from Hassell

Set

$$\varphi_n(y) = \sqrt{\frac{2}{\pi}} \sin(n(y + \pi/2)), \quad y \in (-\pi/2, \pi/2).$$

Then $-\partial_y^2 \varphi_n = n^2 \varphi_n$ and $\varphi_n(\pm\pi/2) = 0$. Fix once and for all a real-valued cutoff

$$\chi \in C_c^\infty((-\pi/4, \pi/4)), \quad \|\chi\|_{L^2(\mathbb{R})} = 1, \quad \int_{\mathbb{R}} \chi(x) dx = 0.$$

For every $t \in [1, 2]$, define

$$v_n(x, y) = \chi(x) \varphi_n(y)$$

in the central rectangle and extend it by zero to S_t . The support of χ is a positive distance away from the two interfaces with the caps, and φ_n satisfies the Dirichlet condition on the horizontal sides. Hence this extension belongs to the domain of the Dirichlet Laplacian and

$$\|v_n\|_{L^2(S_t)} = 1, \quad (H_t - n^2)v_n = (-\chi''(x))\varphi_n(y).$$

Put

$$K_\chi = \|\chi''\|_{L^2(\mathbb{R})}.$$

Then

$$\|(H_t - n^2)v_n\|_{L^2(S_t)} = K_\chi \tag{2.1}$$

uniformly in n and t . Notice that $K_\chi > 0$: otherwise χ would be affine on its support and hence, by compact support, identically zero, contradicting $\|\chi\|_2 = 1$.

Hassell's proof of Theorem 4 in [4] splits the parameter set into two parts, denoted here by Z_1 and Z_2 ; see his equation (9). The following proposition is an extracted corollary of that proof, specialized to the simple stadium family above and to the fixed cutoff χ .

Proposition 2.1 (Hassell input in the form used below). *Let Z_1, Z_2 be the partition of $[1, 2]$ arising in Hassell's proof for the Dirichlet stadium family; thus Z_1 is the set defined by Hassell's condition*

$$\liminf_{j \rightarrow \infty} f_j(t) = 0,$$

and $Z_2 = [1, 2] \setminus Z_1$. Let

$$v_n(x, y) = \chi(x) \varphi_n(y)$$

in the central rectangle and extend it by zero to S_t , as above. Then the following assertions hold.

- (a) *For every $t \in Z_1$ there is a sequence of normalized Dirichlet eigenfunctions w_j of H_t , with eigenvalues tending to infinity, and an associated interior semiclassical measure μ , such that μ gives zero mass to the preimage of the interiors of the two caps under the base projection. In particular, for every*

$$Q_0 \in C_c^\infty(S_t^\circ \setminus \overline{R_t})$$

one has, after passing to the subsequence defining μ ,

$$\langle Q_0 w_j, w_j \rangle_{L^2(S_t)} \longrightarrow 0.$$

In Hassell's formulation, modulo the standard gliding and singular boundary sets, the corresponding quantum limit is supported on the bouncing-ball covectors in the rectangular part.

- (b) *For Lebesgue almost every $t \in Z_2$ there exist integers $n_j \rightarrow \infty$, normalized real Dirichlet eigenfunctions u_j , and constants $C_t, c_t > 0$ such that*

$$H_t u_j = \lambda_j u_j, \quad |\lambda_j - n_j^2| \leq C_t,$$

and

$$|\langle u_j, v_{n_j} \rangle_{L^2(S_t)}| \geq c_t$$

for every j .

Proof. We first prove the Z_1 statement in the precise form needed later. Hassell's Z_1 argument gives a boundary semiclassical measure that vanishes on the curved sides of the stadium. The relation between boundary measures and interior quantum limits then gives an interior quantum limit supported on rays that do not meet the curved sides. For the Bunimovich stadium, the nonsingular interior rays with this property are the bouncing-ball rays in the rectangular part. At the four curvature-jump points of the stadium boundary, we use the clarification of Mangoubi and Weller Weiser [6]; it shows that Hassell's use of the non-gliding-point argument still gives the conclusion needed here: the interior measure obtained in the Z_1 alternative gives no mass to compact subsets of the cap interiors. Hence, for any $Q_0 \in C_c^\infty(S_t^\circ \setminus \overline{R_t})$, multiplication by Q_0 is an interior zeroth-order observable with principal symbol $Q_0 \circ \pi$, where $\pi : S^* S_t^\circ \rightarrow S_t^\circ$ is the base projection. Since $Q_0 \circ \pi$ vanishes on $\text{supp } \mu$,

$$\langle Q_0 w_j, w_j \rangle_{L^2(S_t)} \longrightarrow \int_{S^* S_t^\circ} Q_0(\pi(z, \zeta)) \, d\mu(z, \zeta) = 0.$$

This proves part (a).

We turn to part (b). Recall that

$$\|(H_t - n^2)v_n\|_{L^2(S_t)} = K_\chi$$

uniformly in n and t . Set

$$a_\chi = 2K_\chi > 0.$$

We shall apply Hassell's bounded-window argument with this fixed half-width $a = a_\chi$.

We first record explicitly the consequence of Hassell's parameter estimate that we need. For fixed $a > 0$ and $\varepsilon > 0$, there is a measurable set

$$F_{\varepsilon, a} \subset Z_2$$

such that

$$|Z_2 \setminus F_{\varepsilon, a}| \leq 4\varepsilon,$$

and there is a finite integer $M_{\varepsilon, a}$ with the following property: for every $t \in F_{\varepsilon, a}$ there are infinitely many integers n such that the closed spectral window

$$[n^2 - a, n^2 + a]$$

contains at most $M_{\varepsilon,a}$ eigenvalues of H_t , counted with multiplicity.

Indeed, this is precisely the content of Hassell's equations (10)–(15), but let us spell out the extraction. Since $t \in Z_2$ means $\liminf_j f_j(t) > 0$, Hassell first chooses, for each $\varepsilon > 0$, a set $G_\varepsilon \subset Z_2$ with

$$|G_\varepsilon| \geq |Z_2| - 2\varepsilon,$$

together with constants $c_\varepsilon > 0$ and N_ε , such that

$$t \in G_\varepsilon, \quad j \geq N_\varepsilon \quad \implies \quad f_j(t) \geq \frac{c_\varepsilon}{2}.$$

Using the eigenvalue variation formula and Weyl bounds, Hassell obtains a constant $C_\varepsilon > 0$, independent of a and n , such that, for all sufficiently large n ,

$$\int_{G_\varepsilon} \#\{\lambda \in \text{Spec}(H_t) : |\lambda - n^2| \leq a\} dt \leq C_\varepsilon a.$$

Define

$$M_{\varepsilon,a} = \left\lceil \frac{C_\varepsilon a}{\varepsilon} \right\rceil.$$

By Chebyshev's inequality, for all sufficiently large n the set

$$A_n = \{t \in G_\varepsilon : \#\{\lambda \in \text{Spec}(H_t) : |\lambda - n^2| \leq a\} \leq M_{\varepsilon,a}\}$$

satisfies

$$|A_n| \geq |G_\varepsilon| - \varepsilon \geq |Z_2| - 3\varepsilon.$$

It remains to pass from large measure for each A_n to membership in infinitely many A_n 's. For all sufficiently large integers k , set

$$D_k = \{t \in Z_2 : t \in A_n \text{ for at least } k \text{ distinct integers } n \text{ satisfying } k \leq n < 5k\}.$$

Then

$$\sum_{n=k}^{5k-1} |A_n| \geq 4k(|Z_2| - 3\varepsilon).$$

On the other hand, by the definition of D_k , a point of D_k can be counted at most $4k$ times in this sum, while a point of $Z_2 \setminus D_k$ can be counted at most k times. Hence

$$\sum_{n=k}^{5k-1} |A_n| \leq 4k|D_k| + k(|Z_2| - |D_k|).$$

Combining the last two estimates gives

$$|D_k| \geq |Z_2| - 4\varepsilon.$$

Since

$$D_k \subset \bigcup_{n \geq k} A_n,$$

we get, for all sufficiently large k ,

$$\left| \bigcup_{n \geq k} A_n \right| \geq |Z_2| - 4\varepsilon.$$

By continuity of measure from above,

$$\left| \limsup_{n \rightarrow \infty} A_n \right| = \left| \bigcap_{k=1}^{\infty} \bigcup_{n \geq k} A_n \right| \geq |Z_2| - 4\varepsilon.$$

Taking

$$F_{\varepsilon, a} = \limsup_{n \rightarrow \infty} A_n$$

proves the bounded-window consequence stated above.

Now choose a sequence $\varepsilon_m \downarrow 0$, for instance

$$\varepsilon_m = 2^{-m},$$

and define

$$Z_2^{\text{good}} = \bigcup_{m=1}^{\infty} F_{\varepsilon_m, a_\chi}.$$

Since

$$|Z_2 \setminus F_{\varepsilon_m, a_\chi}| \leq 4\varepsilon_m$$

for every m , we have

$$|Z_2 \setminus Z_2^{\text{good}}| = \left| \bigcap_{m=1}^{\infty} (Z_2 \setminus F_{\varepsilon_m, a_\chi}) \right| \leq \inf_m 4\varepsilon_m = 0.$$

Thus Z_2^{good} has full measure in Z_2 .

Fix $t \in Z_2^{\text{good}}$. Then $t \in F_{\varepsilon_m, a_\chi}$ for some m . Let

$$M_t = M_{\varepsilon_m, a_\chi}.$$

There are infinitely many integers n such that

$$\#\{\lambda \in \text{Spec}(H_t) : |\lambda - n^2| \leq a_\chi\} \leq M_t.$$

Choose such an n . Let

$$P_n = \mathbf{1}_{(n^2 - a_\chi, n^2 + a_\chi)}(H_t)$$

be the spectral projection onto the open window. Its rank is at most M_t , since the open window is contained in the closed window counted above. On the complementary spectral subspace,

$$|\lambda - n^2| \geq a_\chi.$$

Therefore, by the spectral theorem and the quasimode estimate,

$$a_\chi \|(I - P_n)v_n\|_{L^2(S_t)} \leq \|(H_t - n^2)(I - P_n)v_n\|_{L^2(S_t)} \leq \|(H_t - n^2)v_n\|_{L^2(S_t)} = K_\chi.$$

Since $a_\chi = 2K_\chi$, this gives

$$\|(I - P_n)v_n\|_{L^2(S_t)} \leq \frac{1}{2},$$

and hence

$$\|P_nv_n\|_{L^2(S_t)}^2 = 1 - \|(I - P_n)v_n\|_{L^2(S_t)}^2 \geq \frac{3}{4}.$$

Let

$$\{u_{n,k}\}_{k=1}^{r_n}$$

be an orthonormal real eigenbasis of $\text{ran } P_n$. Such a real basis may be chosen because H_t has real coefficients and the Dirichlet condition is real. Since $r_n \leq M_t$,

$$\sum_{k=1}^{r_n} |\langle u_{n,k}, v_n \rangle_{L^2(S_t)}|^2 = \|P_nv_n\|_{L^2(S_t)}^2 \geq \frac{3}{4}.$$

Consequently, for at least one k ,

$$|\langle u_{n,k}, v_n \rangle_{L^2(S_t)}| \geq \sqrt{\frac{3}{4M_t}}.$$

Selecting one such eigenfunction for each of the infinitely many admissible integers n , and relabelling them as u_j with $n = n_j \rightarrow \infty$, we obtain

$$H_t u_j = \lambda_j u_j, \quad |\lambda_j - n_j^2| < a_\chi,$$

and

$$|\langle u_j, v_{n_j} \rangle_{L^2(S_t)}| \geq \sqrt{\frac{3}{4M_t}}.$$

Thus part (b) holds with, for example,

$$C_t = a_\chi = 2K_\chi, \quad c_t = \sqrt{\frac{3}{4M_t}}.$$

The proof is complete. $\square$

Remark 2.2. *Proposition 2.1 is where Hassell's parameter argument enters. Proposition 4.1 below is purely local to the central rectangle and may be read independently as the following implication: fixed bounded-window overlap with one bouncing-ball quasimode already implies failure of physical equidistribution.*

3 One-dimensional lemmas in the rectangle

We use three elementary facts about solutions of one-dimensional constant-coefficient equations. They are stated with constants uniform over the compact parameter ranges needed later.

Lemma 3.1 (Compactness of bounded-frequency solutions). *Let $I = (-T, T)$ and let μ_j be a bounded sequence of real numbers. If*

$$-b_j'' = \mu_j b_j, \quad \|b_j\|_{L^2(I)} = 1,$$

then a subsequence converges in $C^1(\bar{I})$ and in $L^2(I)$ to a non-zero solution b_∞ of

$$-b''_\infty = \mu_\infty b_\infty$$

for some subsequential limit μ_∞ .

Proof. After passing to a subsequence, assume $\mu_j \rightarrow \mu_\infty$. It remains to obtain compactness. Choose C with $|\mu_j| \leq C$. For each $|\mu| \leq C$ and each initial vector $\xi = (\xi_0, \xi_1) \in \mathbb{R}^2$, let $B_{\mu, \xi}$ be the solution of

$$-B'' = \mu B, \quad B(0) = \xi_0, \quad B'(0) = \xi_1.$$

The quadratic form

$$G_\mu(\xi) = \|B_{\mu, \xi}\|_{L^2(I)}^2$$

is positive definite for every μ , because a non-zero solution of a second-order ODE cannot vanish identically on an interval. Since $\mu \mapsto G_\mu$ is continuous on the compact interval $[-C, C]$, there is $c_0 > 0$ such that

$$G_\mu(\xi) \geq c_0 |\xi|^2 \quad (|\mu| \leq C).$$

Thus the initial data $(b_j(0), b'_j(0))$ are uniformly bounded. The ODE then gives uniform bounds for b_j , b'_j , and b''_j on $\bar{I}$. Arzela–Ascoli gives a subsequence converging in $C^1(\bar{I})$ to a solution of the limiting equation. The L^2 norm of the limit is one, so the limit is non-zero. $\square$

Lemma 3.2 (Rapid oscillation). *Let $q \in C_c^\infty(I)$ with $\int_I q(x) dx = 0$. There is a constant C_q , depending on q and I , such that every real solution of*

$$-a'' = k^2 a, \quad k \geq 1,$$

satisfies

$$\left| \int_I q(x) |a(x)|^2 dx \right| \leq \frac{C_q}{k} \|a\|_{L^2(I)}^2.$$

Proof. Write

$$a(x) = A \cos(kx) + B \sin(kx).$$

Then

$$|a(x)|^2 = \frac{A^2 + B^2}{2} + \frac{A^2 - B^2}{2} \cos(2kx) + AB \sin(2kx).$$

The constant term integrates to zero against q . One integration by parts gives

$$\left| \int_I q(x) e^{2ikx} dx \right| \leq \frac{\|q'\|_{L^1(I)}}{2k}.$$

Therefore

$$\left| \int_I q(x) |a(x)|^2 dx \right| \leq \frac{C'_q}{k} (A^2 + B^2).$$

It remains to compare $A^2 + B^2$ with $\|a\|_2^2$ uniformly for $k \geq 1$. For $1 \leq k \leq K$, set

$$m_K = \min_{\substack{1 \leq k \leq K \\ A^2 + B^2 = 1}} \|A \cos(kx) + B \sin(kx)\|_{L^2(I)}^2.$$

The parameter set is compact, and no non-zero function $A \cos(kx) + B \sin(kx)$ can vanish identically on I , so $m_K > 0$. This gives the comparison for $1 \leq k \leq K$. For $k \geq K$ it follows directly from the identities for the integrals of $\cos^2(kx)$, $\sin^2(kx)$, and $\sin(kx) \cos(kx)$: the corresponding Gram matrix is T times the identity plus an $O(k^{-1})$ matrix. Taking K large enough gives

$$A^2 + B^2 \leq C_I \|a\|_{L^2(I)}^2$$

for all $k \geq 1$, and the lemma follows. $\square$

Lemma 3.3 (Evanescent modes in the interior). *Let $J \Subset I = (-T, T)$, meaning that the closure of J is contained in I . There are constants $C, c > 0$, depending only on I and J , such that every real solution of*

$$a'' = \alpha^2 a, \quad \alpha \geq 1,$$

satisfies

$$\|a\|_{L^2(J)} \leq C e^{-c\alpha} \|a\|_{L^2(I)}.$$

Proof. Choose $\delta > 0$ such that

$$J \subset [-T + \delta, T - \delta].$$

Every solution has the form

$$a(x) = P_+ e^{\alpha(x-T)} + P_- e^{-\alpha(x+T)}.$$

On J both exponentials are $O(e^{-\alpha\delta})$, and hence

$$\|a\|_{L^2(J)}^2 \leq C_J e^{-2\alpha\delta} (|P_+|^2 + |P_-|^2). \quad (3.1)$$

On I , let

$$f_1(x) = e^{\alpha(x-T)}, \quad f_2(x) = e^{-\alpha(x+T)}.$$

The Gram matrix of f_1 and f_2 has diagonal entries

$$\langle f_1, f_1 \rangle = \langle f_2, f_2 \rangle = \frac{1 - e^{-4\alpha T}}{2\alpha}$$

and off-diagonal entry

$$\langle f_1, f_2 \rangle = 2T e^{-2\alpha T}.$$

The diagonal entries are comparable to α^{-1} for $\alpha \geq 1$, while the off-diagonal entry is exponentially small. For large α this gives a lower bound of order α^{-1} for the least eigenvalue, and the remaining bounded range of α is handled by compactness and linear independence of f_1, f_2 . Hence, after decreasing the constant if necessary,

$$\|a\|_{L^2(I)}^2 \geq c_0 \alpha^{-1} (|P_+|^2 + |P_-|^2), \quad \alpha \geq 1. \quad (3.2)$$

Combining (3.1) and (3.2) gives

$$\|a\|_{L^2(J)}^2 \leq C \alpha e^{-2\alpha\delta} \|a\|_{L^2(I)}^2.$$

Choose $0 < c < 2\delta$. Since $\alpha e^{-(2\delta-c)\alpha}$ is bounded for $\alpha \geq 1$, we obtain

$$\|a\|_{L^2(J)}^2 \leq C e^{-c\alpha} \|a\|_{L^2(I)}^2.$$

Replacing c by $c/2$ and adjusting C gives the stated norm estimate. $\square$

4 Physical extraction from a fixed vertical overlap

This section is the elementary core of the paper. It is independent of Hassell's parameter argument.

Proposition 4.1 (Fixed-overlap extraction). *Fix $t \in [1, 2]$. Let u_j be normalized real Dirichlet eigenfunctions on S_t and let $n_j \rightarrow \infty$ be integers such that*

$$H_t u_j = \lambda_j u_j, \quad |\lambda_j - n_j^2| \leq C,$$

and

$$|\langle u_j, \chi \varphi_{n_j} \rangle_{L^2(S_t)}| \geq c > 0.$$

Then, after passing to a subsequence, there exists a real-valued $Q \in C^\infty(\overline{S}_t)$ with

$$\int_{S_t} Q(z) dz = 0$$

such that

$$\langle Q u_j, u_j \rangle_{L^2(S_t)} \longrightarrow L \neq 0.$$

Proof. All arguments in this proof are local to the rectangle R_t . For $\ell \geq 1$, define the vertical Fourier coefficient

$$a_{j,\ell}(x) = \int_{-\pi/2}^{\pi/2} u_j(x, y) \varphi_\ell(y) dy, \quad x \in I_t.$$

Since u_j and φ_ℓ are real-valued, all $a_{j,\ell}$ are real-valued. Since u_j solves $(-\partial_x^2 - \partial_y^2)u_j = \lambda_j u_j$ in R_t and satisfies the Dirichlet condition on the horizontal sides $y = \pm\pi/2$, testing the equation against functions of the form $\eta(x)\varphi_\ell(y)$, with $\eta \in C_c^\infty(I_t)$, gives, in the distributional sense on I_t ,

$$-a_{j,\ell}''(x) + \ell^2 a_{j,\ell}(x) = \lambda_j a_{j,\ell}(x).$$

Equivalently,

$$-a_{j,\ell}'' = (\lambda_j - \ell^2) a_{j,\ell} \quad \text{on } I_t. \tag{4.1}$$

Because the right-hand side belongs to $L^2(I_t)$, the coefficient belongs to $H_{\text{loc}}^2(I_t)$, and the constant-coefficient ODE then implies smoothness on I_t . No boundary condition at the endpoints of I_t is used; the caps affect how the coefficients match to the exterior of the central rectangle.

By Parseval in the vertical variable,

$$\sum_{\ell \geq 1} \|a_{j,\ell}\|_{L^2(I_t)}^2 = \|u_j\|_{L^2(R_t)}^2 \leq 1. \tag{4.2}$$

The overlap assumption is exactly

$$\left| \int_{I_t} a_{j,n_j}(x) \chi(x) dx \right| \geq c. \tag{4.3}$$

Let

$$A_j = \|a_{j,n_j}\|_{L^2(I_t)}.$$

Since $\|\chi\|_2 = 1$, (4.3) and Cauchy's inequality give

$$c \leq A_j \leq 1.$$

Passing to a subsequence, assume

$$A_j \rightarrow A_\infty \in [c, 1], \quad \mu_j := \lambda_j - n_j^2 \rightarrow \mu_\infty.$$

Set

$$b_j = A_j^{-1} a_{j, n_j}.$$

Then

$$\|b_j\|_{L^2(I_t)} = 1, \quad -b_j'' = \mu_j b_j.$$

By Lemma 3.1, after passing to a further subsequence,

$$b_j \rightarrow b_\infty$$

in $C^1(\bar{I}_t)$ and in $L^2(I_t)$, where $\|b_\infty\|_2 = 1$ and $-b_\infty'' = \mu_\infty b_\infty$. Moreover, the L^2 convergence gives

$$\left| \int_{I_t} b_\infty(x) \chi(x) dx \right| = \lim_{j \rightarrow \infty} A_j^{-1} \left| \int_{I_t} a_{j, n_j}(x) \chi(x) dx \right| \geq c.$$

Since $\int_{\mathbb{R}} \chi(x) dx = 0$, the limit b_∞ is not constant. The functions are real-valued. A nonconstant real solution of a constant-coefficient second-order ODE cannot have constant square on the connected interval I_t : if b_∞^2 were constant, then b_∞ would have constant absolute value, hence by continuity a constant sign and therefore would itself be constant. Thus b_∞^2 is not constant.

We now choose a detector in the horizontal variable. There exists a real-valued $q \in C_c^\infty(I_t)$ such that

$$\int_{I_t} q(x) dx = 0, \quad \int_{I_t} q(x) b_\infty(x)^2 dx \neq 0. \quad (4.4)$$

Indeed, choose $\rho \in C_c^\infty(I_t)$ with $\int_{I_t} \rho(x) dx = 1$. If the second integral vanished for every real-valued test function $q \in C_c^\infty(I_t)$ with zero integral, then for every $\psi \in C_c^\infty(I_t)$ the test function $\psi - (\int_{I_t} \psi(x) dx) \rho$ would give

$$\int_{I_t} \psi(x) b_\infty(x)^2 dx = \left(\int_{I_t} \psi(x) dx \right) \int_{I_t} \rho(x) b_\infty(x)^2 dx.$$

Thus b_∞^2 would be constant as a distribution on I_t , contradicting the previous paragraph.

Extend q by zero to a function in $C_c^\infty(\mathbb{R})$, still denoted by q , and define the physical observable by

$$Q(x, y) = q(x)$$

in a neighbourhood of $\bar{S}_t$. Since $\text{supp } q \Subset I_t$, this function vanishes near the two cap interfaces and on the caps, and its restriction gives $Q \in C^\infty(\bar{S}_t)$. Also

$$\int_{S_t} Q(z) dz = \pi \int_{I_t} q(x) dx = 0. \quad (4.5)$$

Because Q is independent of y in R_t and vanishes on $S_t \setminus R_t$, orthogonality of the vertical basis gives

$$\langle Qu_j, u_j \rangle_{L^2(S_t)} = \sum_{\ell=1}^{\infty} \int_{I_t} q(x) |a_{j,\ell}(x)|^2 dx. \quad (4.6)$$

The identity follows first for finite vertical sums and then by L^2 convergence and boundedness of q .

We separate the target mode, the lower vertical modes, and the higher vertical modes. For the target mode, the L^2 convergence $b_j \rightarrow b_\infty$ gives

$$\left| \int_{I_t} q(b_j^2 - b_\infty^2) dx \right| \leq \|q\|_\infty \|b_j - b_\infty\|_2 (\|b_j\|_2 + \|b_\infty\|_2) \rightarrow 0.$$

Therefore

$$\int_{I_t} q |a_{j,n_j}|^2 = A_j^2 \int_{I_t} q b_j^2 \rightarrow A_\infty^2 \int_{I_t} q b_\infty^2 =: L. \quad (4.7)$$

By (4.4) and $A_\infty \geq c > 0$, $L \neq 0$.

All constants in the following mode estimates may depend on the fixed data t and q , and later on the chosen interval J , but not on j or ℓ .

For $\ell < n_j$, put

$$k_{j,\ell} = \sqrt{\lambda_j - \ell^2}.$$

Since $\lambda_j = n_j^2 + O(1)$,

$$k_{j,\ell}^2 \geq n_j^2 - C - (n_j - 1)^2 = 2n_j - 1 - C$$

for all $\ell < n_j$ and all large j . Thus $k_{j,\ell} \geq c_0 \sqrt{n_j}$, in particular $k_{j,\ell} \geq 1$. By (4.1) and Lemma 3.2,

$$\left| \int_{I_t} q |a_{j,\ell}|^2 \right| \leq C_q n_j^{-1/2} \|a_{j,\ell}\|_{L^2(I_t)}^2.$$

Summing and using (4.2),

$$\left| \sum_{\ell < n_j} \int_{I_t} q |a_{j,\ell}|^2 \right| \leq C_q n_j^{-1/2} \sum_{\ell < n_j} \|a_{j,\ell}\|_{L^2(I_t)}^2 \leq C_q n_j^{-1/2} \rightarrow 0. \quad (4.8)$$

For $\ell > n_j$, put

$$\alpha_{j,\ell} = \sqrt{\ell^2 - \lambda_j}.$$

Again $\lambda_j = n_j^2 + O(1)$ gives, for all large j ,

$$\alpha_{j,\ell}^2 \geq (n_j + 1)^2 - n_j^2 - C = 2n_j + 1 - C,$$

and hence $\alpha_{j,\ell} \geq c_0 \sqrt{n_j}$. Choose an interval $J \Subset I_t$ with $\text{supp } q \subset J$. By (4.1), the coefficient satisfies $a_{j,\ell}'' = \alpha_{j,\ell}^2 a_{j,\ell}$. Therefore Lemma 3.3 gives, after squaring the estimate and renaming constants,

$$\|a_{j,\ell}\|_{L^2(J)}^2 \leq C e^{-c\alpha_{j,\ell}} \|a_{j,\ell}\|_{L^2(I_t)}^2 \leq C e^{-c' \sqrt{n_j}} \|a_{j,\ell}\|_{L^2(I_t)}^2.$$

Consequently,

$$\begin{aligned} \left| \sum_{\ell > n_j} \int_{I_t} q |a_{j,\ell}|^2 \right| &\leq \|q\|_\infty \sum_{\ell > n_j} \|a_{j,\ell}\|_{L^2(J)}^2 \\ &\leq C e^{-c' \sqrt{n_j}} \sum_{\ell > n_j} \|a_{j,\ell}\|_{L^2(I_t)}^2 \leq C e^{-c' \sqrt{n_j}} \rightarrow 0. \end{aligned} \quad (4.9)$$

Combining (4.6), (4.7), (4.8), and (4.9) yields

$$\langle Qu_j, u_j \rangle_{L^2(S_t)} \longrightarrow L \neq 0.$$

Together with (4.5), this proves the proposition. $\square$

Remark 4.2. *The observable Q is chosen after passing to the subsequence on which $b_j \rightarrow b_\infty$. Once chosen, it is fixed for all members of that subsequence. This is exactly what is required to contradict physical equidistribution along that subsequence.*

5 From a smooth observable to a region

Lemma 5.1 (Level-set extraction). *Let $\Omega \subset \mathbb{R}^2$ be a bounded Lebesgue-measurable set with $0 < |\Omega| < \infty$, let ν_j be probability measures on Ω , and let $m_\Omega = |\Omega|^{-1} \mathbf{1}_\Omega dz$. Let Q be a bounded real-valued Borel function on Ω whose restriction to Ω° is smooth. If*

$$\int_\Omega Q d\nu_j \not\rightarrow \int_\Omega Q dm_\Omega,$$

then there is a regular value s of $Q|_{\Omega^\circ}$ such that, for

$$A_s = \{z \in \Omega : Q(z) > s\},$$

one has

$$\nu_j(A_s) \not\rightarrow m_\Omega(A_s).$$

The relative boundary of $A_s \cap \Omega^\circ$ in Ω° is

$$\partial_{\Omega^\circ}(A_s \cap \Omega^\circ) = \{Q = s\} \cap \Omega^\circ,$$

and is smooth. If, in addition, all ν_j are absolutely continuous with respect to Lebesgue measure, as in the eigenfunction application below, then A_s may be modified on Lebesgue-null subsets of the ambient plane, in particular on boundary pieces, without changing any of the masses.

Proof. Choose M so that $|Q(z)| < M$ for every $z \in \Omega$. For every probability measure ν on Ω ,

$$\int_\Omega Q d\nu = -M + \int_{-M}^M \nu(\{Q > s\}) ds. \quad (5.1)$$

This is the standard layer-cake identity for a bounded real-valued random variable.

Suppose, to the contrary, that

$$\nu_j(\{Q > s\}) \rightarrow m_\Omega(\{Q > s\})$$

for every regular value s of $Q|_{\Omega^\circ}$. By Sard's theorem, the set of critical values has Lebesgue measure zero. Hence the pointwise convergence above holds for almost every $s \in (-M, M)$. Since all terms are bounded by 1, dominated convergence in (5.1) gives

$$\int_\Omega Q d\nu_j \rightarrow \int_\Omega Q dm_\Omega,$$

contradiction. Thus some regular value s gives non-convergence of the superlevel-set masses. The smoothness of the interior level set follows from the regular-value theorem. $\square$

6 Proof of the main theorem

Proof of Theorem 1.1. Let $Z_2^{\text{good}} \subset Z_2$ be the full-measure subset on which Proposition 2.1(b) holds, and set

$$\mathcal{G} = Z_1 \cup Z_2^{\text{good}}.$$

Since $Z_1 \cup Z_2 = [1, 2]$, the set $\mathcal{G}$ has full measure in $[1, 2]$. Fix $t \in \mathcal{G}$.

First suppose $t \in Z_1$. By Proposition 2.1(a), there is a normalized eigenfunction sequence w_j such that

$$\langle Q_0 w_j, w_j \rangle_{L^2(S_t)} \rightarrow 0$$

for every $Q_0 \in C_c^\infty(S_t^\circ \setminus \overline{R_t})$. Choose a non-negative function Q_0 , not identically zero, supported compactly in the interiors of the two caps, and extend Q_0 by zero to S_t . Since Q_0 is supported away from the cap interfaces and the billiard boundary, this extension is smooth on $\overline{S_t}$. Thus

$$\langle Q_0 w_j, w_j \rangle \rightarrow 0.$$

To obtain real eigenfunctions, write $w_j = v_j + ir_j$ with v_j, r_j real. Since the Dirichlet Laplacian is real, both v_j and r_j are eigenfunctions. Choose $h_j \in \{v_j, r_j\}$ with $\|h_j\|_2 \geq 1/\sqrt{2}$ and set $u_j = h_j/\|h_j\|_2$. Then u_j is a normalized real eigenfunction and, since $Q_0 \geq 0$,

$$\langle Q_0 u_j, u_j \rangle \leq 2\langle Q_0 w_j, w_j \rangle \rightarrow 0.$$

On the other hand,

$$\frac{1}{|S_t|} \int_{S_t} Q_0(z) dz > 0.$$

Thus physical equidistribution fails. To obtain the mean-zero observable asserted in the theorem, set

$$Q = Q_0 - \frac{1}{|S_t|} \int_{S_t} Q_0(z) dz.$$

Then $\int_{S_t} Q = 0$ and

$$\langle Qu_j, u_j \rangle \rightarrow -\frac{1}{|S_t|} \int_{S_t} Q_0(z) dz \neq 0.$$

Since $\int_{S_t} Q \, dm_{S_t} = 0$, applying [Lemma 5.1](#) to the probability measures $|u_j|^2 dz$ and to $m_{S_t} = |S_t|^{-1} dz$ gives a physical region $A \subset S_t$ with smooth interior relative boundary such that

$$\int_A |u_j|^2 dz \not\rightarrow \frac{|A|}{|S_t|}.$$

This proves the theorem in the Z_1 case.

Now suppose $t \in Z_2^{\text{good}}$. Then there are eigenfunctions u_j and integers $n_j \rightarrow \infty$ satisfying the hypotheses of [Proposition 4.1](#). Applying that proposition, after passing to a subsequence we obtain a smooth real-valued mean-zero observable Q with

$$\langle Qu_j, u_j \rangle \rightarrow L \neq 0.$$

Since $\int_{S_t} Q \, dz = 0$, this contradicts physical equidistribution. Applying [Lemma 5.1](#) to the probability measures $|u_j|^2 dz$ and to $m_{S_t} = |S_t|^{-1} dz$ gives a physical region $A \subset S_t$ with smooth interior relative boundary such that

$$\int_A |u_j|^2 dz \not\rightarrow \frac{|A|}{|S_t|}.$$

This proves the theorem in the Z_2 case as well. $\square$

Remark 6.1. *The argument proves the physical-space statement but not the stronger assertion*

$$\int_{R_t} |u_j|^2 \rightarrow 1.$$

In the bounded-window case, localization can be distributed across a finite cluster in phase space. [Proposition 4.1](#) avoids this by using the exact vertical Fourier coefficient selected by the fixed bouncing-ball overlap.

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Declaration on the use of AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition.

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